

Chapter 6

The Link Layer and LANs

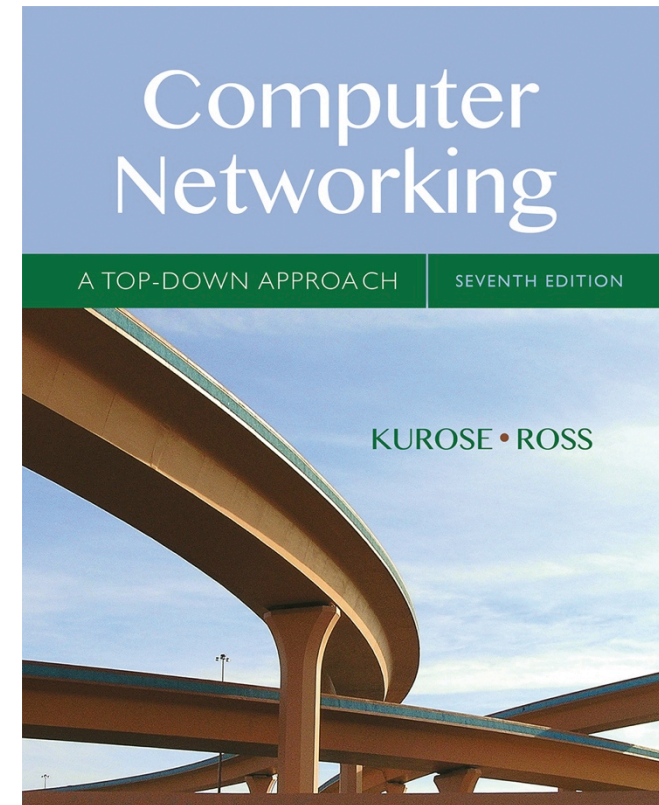
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Thanks and enjoy! JFK/KWR

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Computer Networking: A Top Down Approach

7th edition

Jim Kurose, Keith Ross

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Link layer, LANs: outline

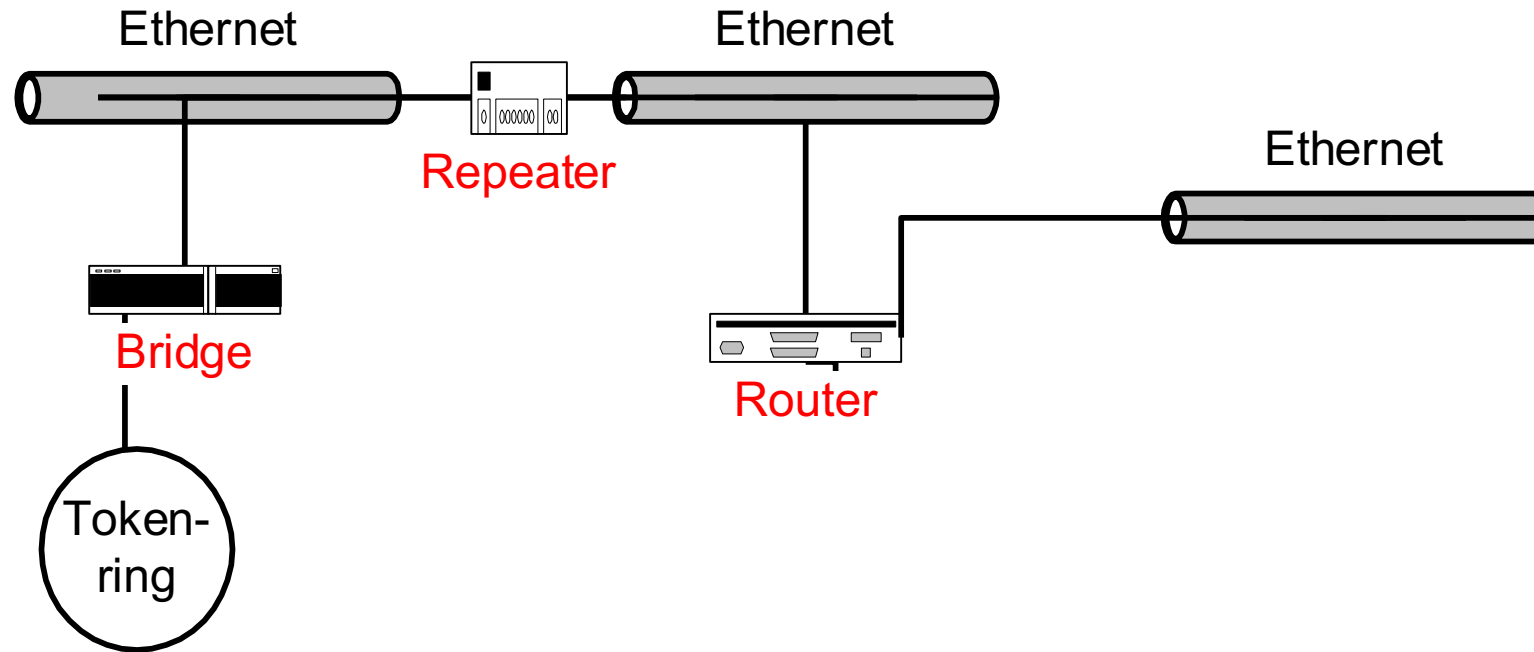
6.4 LAN Switches

- Interconnection devices – Repeaters, Bridges¹, Routers
- Bridges/LAN switches vs. Routers
- Bridges
- Learning Bridges
- Transparent Bridges

¹bridge: old name for switch

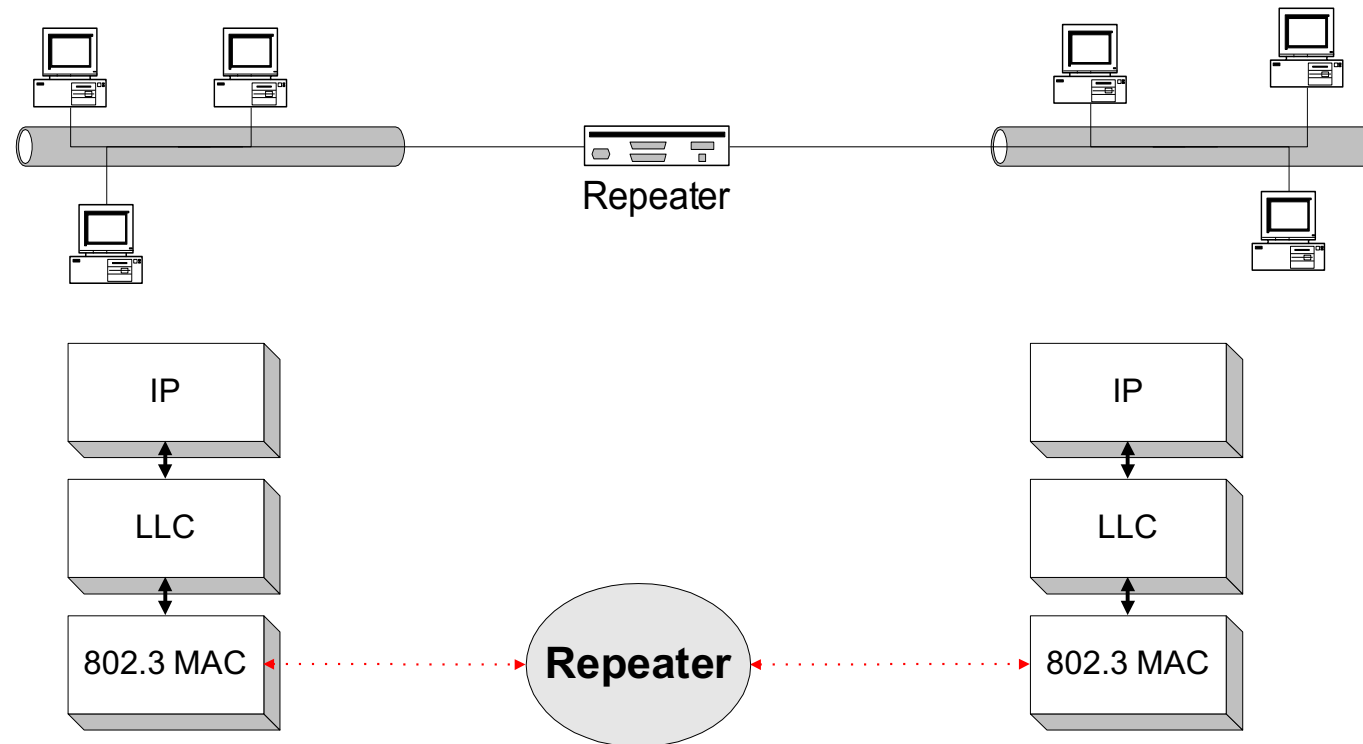
Interconnection Devices

- there are many different devices for interconnecting networks.



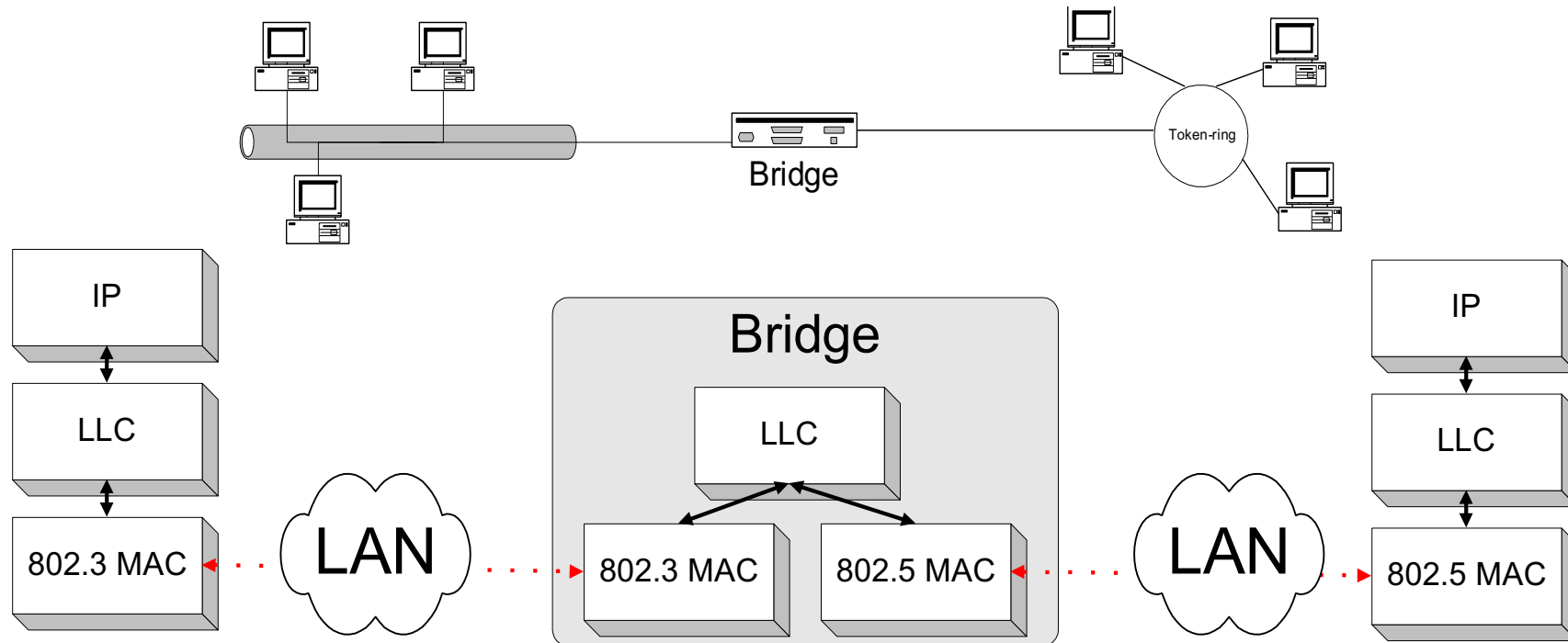
Repeaters

- used to interconnect multiple Ethernet segments
- merely extends the baseband cable
- amplifies all signals including collisions/errors



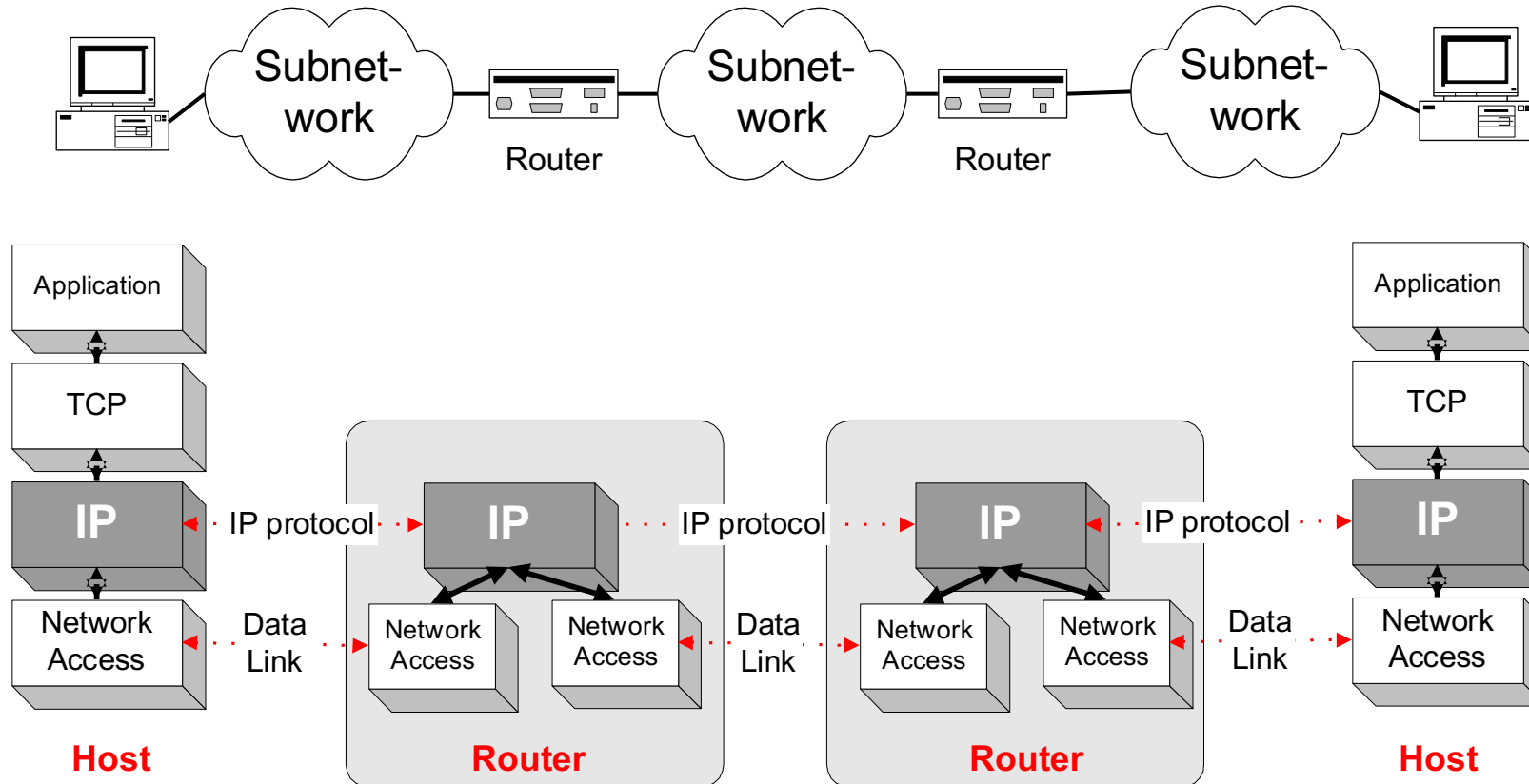
Bridges/LAN switches

- interconnect multiple LANs, possibly different types
- bridges operate at the Data Link Layer (Layer 2) and only forward (switch) link layer frames
- forwarding is done based on MAC addresses and hop-by-hop



Routers

- routers operate at the Network Layer (Layer 3)
- interconnect different subnetworks
- use a forwarding table and IP addresses to route packets hop-by-hop

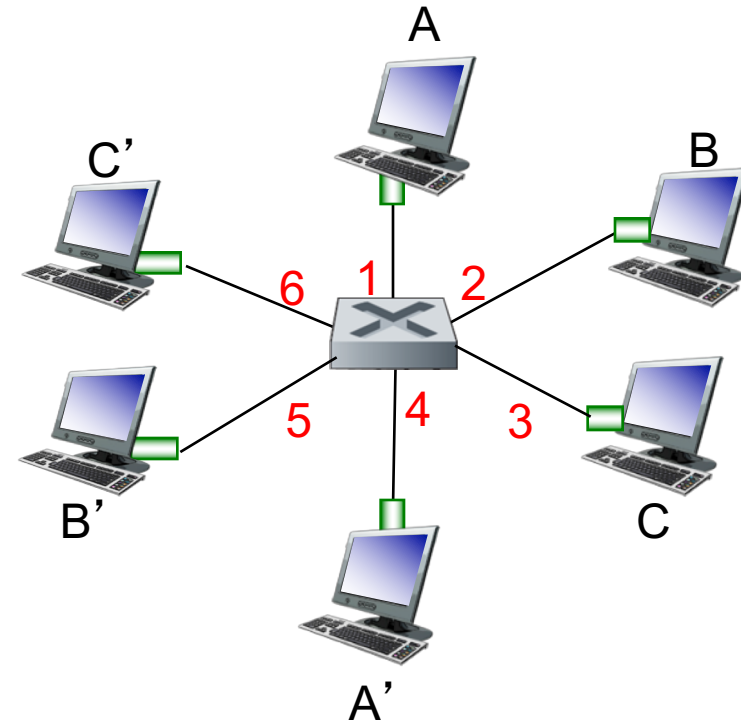


Ethernet switch

- **link-layer device: takes an *active* role in forwarding frames**
 - store, forward Ethernet frames
 - examine incoming frame's MAC address,
 - ***selectively*** forward frame to one-or-more outgoing links
 - when frame is to be forwarded on segment, uses CSMA/CD to access segment
- ***transparent***
 - hosts are unaware of presence of switches
- ***plug-and-play, self-learning***
 - switches do not need to be configured, i.e., they create and manage their own forwarding tables

Switch: *multiple* simultaneous transmissions

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, but no collisions; full duplex
 - each link is its own collision domain
- **switching**: A-to-A' and B-to-B' can transmit simultaneously, without collisions



*switch with six interfaces
(1,2,3,4,5,6)*

Switch forwarding table

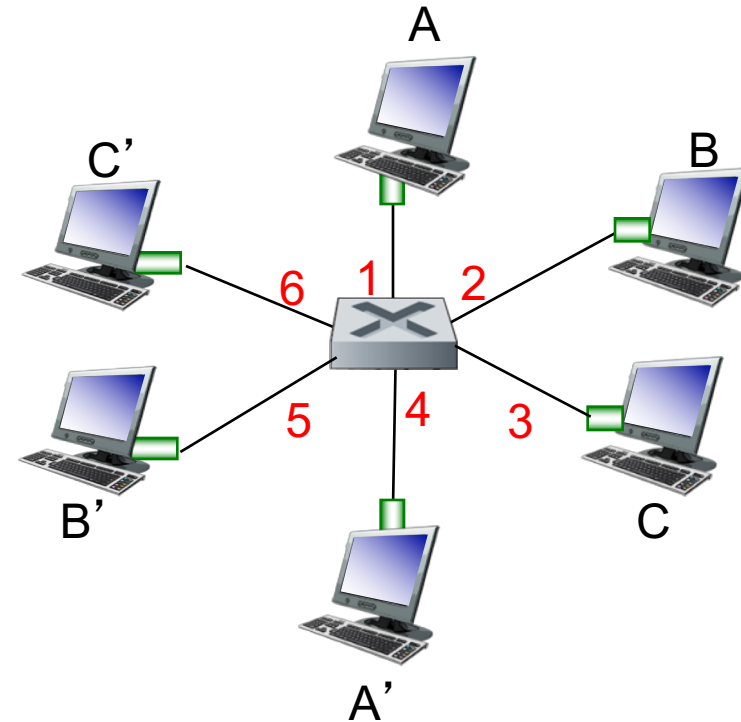
Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

A: each switch has a forwarding (**switch**) table, each entry consist of:

- MAC address of host, interface to reach host, time stamp
- looks like a routing table!

Q: how are entries created, maintained in forwarding table?

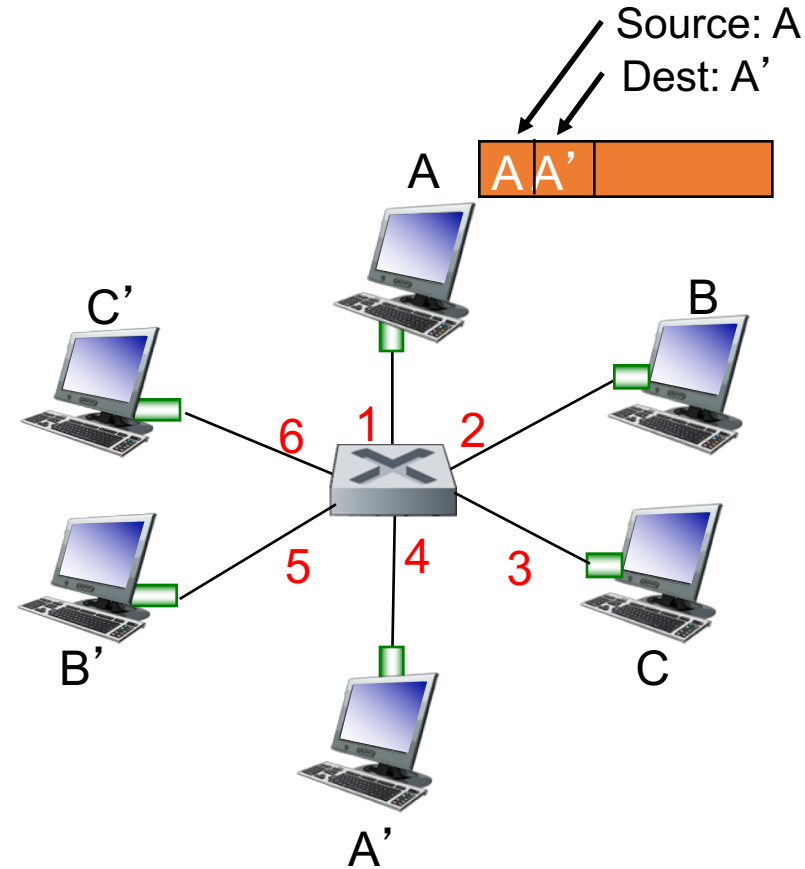
- uses a backwards learning algorithm (something like a routing protocol) to populate table with forwarding entries



*switch with six interfaces
(1,2,3,4,5,6)*

Switch: self-learning

- switch *learns* which hosts can be reached through which interfaces
 - when frame received, switch “learns” location of sender: incoming LAN segment
 - records sender/location pair in switch table

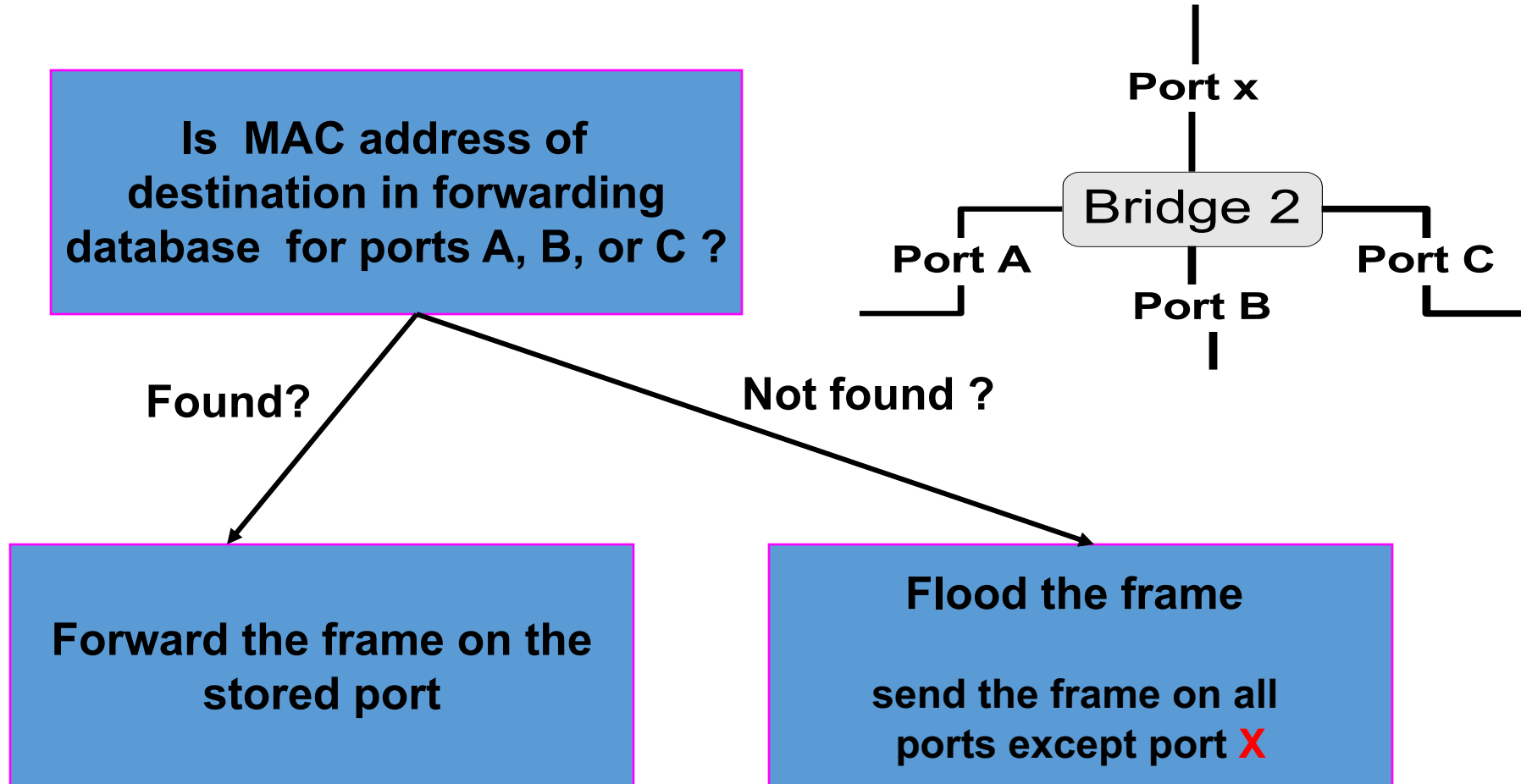


MAC addr	interface	TTL
A	1	60

*Switch table
(initially empty)*

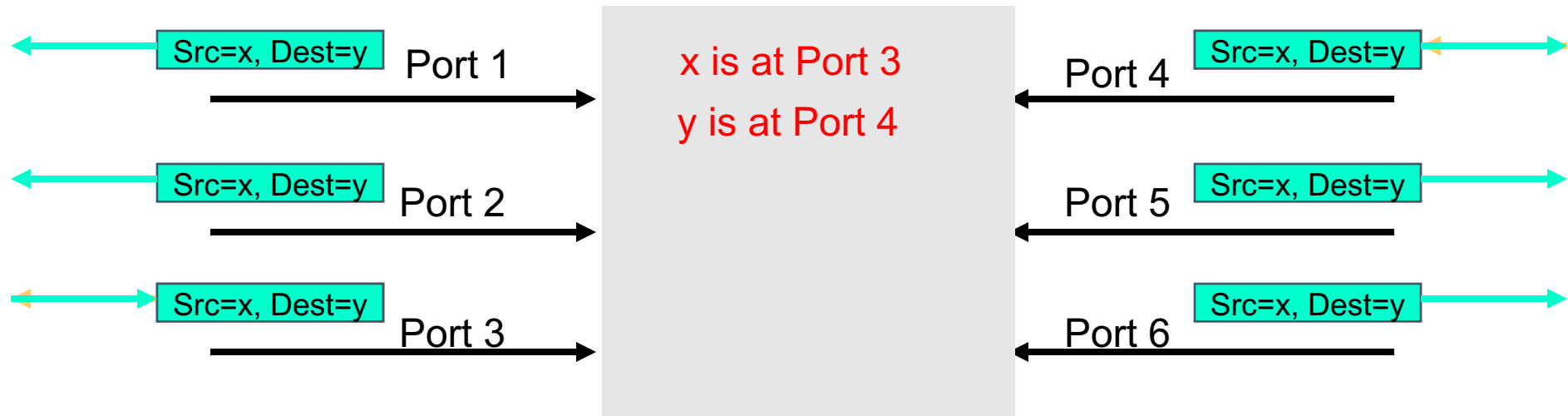
Frame Forwarding

- assume a MAC frame arrives on port **X**



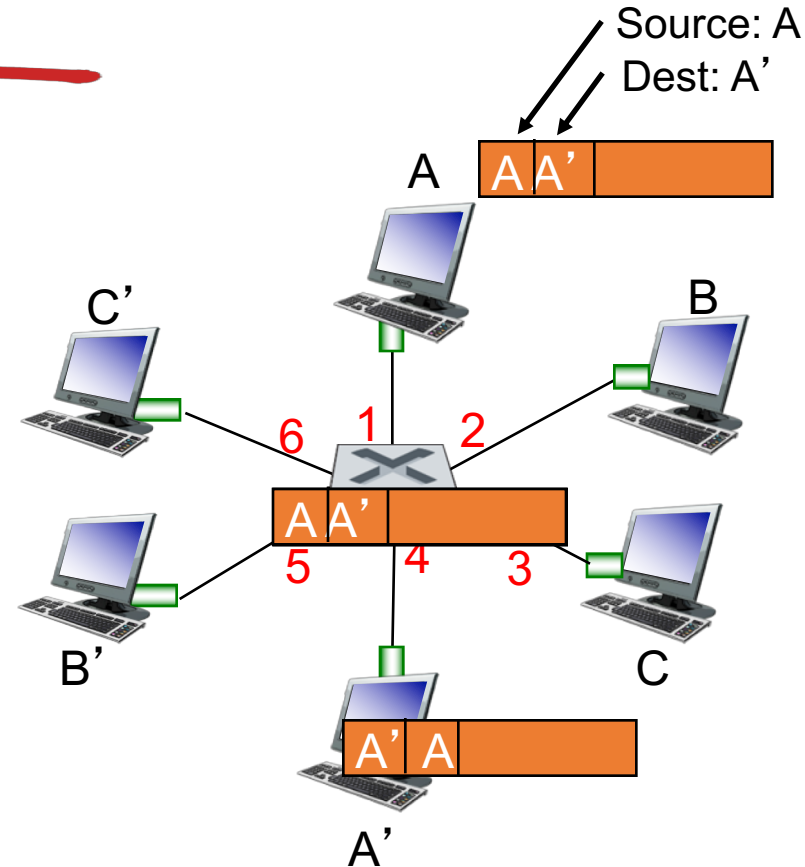
Self Learning (Backwards Learning)

- routing tables entries are set automatically with a simple heuristic:
 - the source address field of a frame that arrives on a port informs switch that this address (host) is reachable from this port



Self-learning, forwarding: example

- frame destination, A' ,
location unknown: *flood*
- destination A location
known: *selectively send*
on just one link



MAC addr	interface	TTL
A	1	60
A'	4	60

switch table
(initially empty)

Learning Bridges Algorithm

For each frame received:

- the bridge enters the source mac address and port in forwarding table or refreshes timer of an existing entry (address seen previously and not yet expired)
- the bridge looks to find entry for destination MAC address in forwarding database
 - if port/interface on which frame is received is same as that for destination, drop frame (e.g., broadcast environment, see next slide)
 - if an entry for destination MAC address exists in forwarding table, reset timer and forward the frame
 - if entry not found, the bridge floods all ports with the frame except for port on which the frame was received
- all entries are deleted after some time (default is 15 seconds).